

CoRoT-1b

R-2596

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_170106 · created 2026-08-27T23:55:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457759.8562 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.042
Transit depth (Rp/Rs)^2	2.02 +/- 1.18 [%]
Orbital Inclination (inc)	89.01 +/- 2.85
Airmass coefficient 1 (a1)	1.92 +/- 1.06
Airmass coefficient 2 (a2)	-0.52 +/- 0.19
Scatter in the residuals of the lightcurve fit is	51.46 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	6.0
Transit Duration (day)	0.1119 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.042 vs 0.1388 (deviation 0.08σ)
Duration fit vs published	0.1119 d vs 0.1046 d (deviation 1.2σ)
Mid-time O–C	5.7 min
Depth SNR	3.4
Residual scatter	51.46 %

Light curve

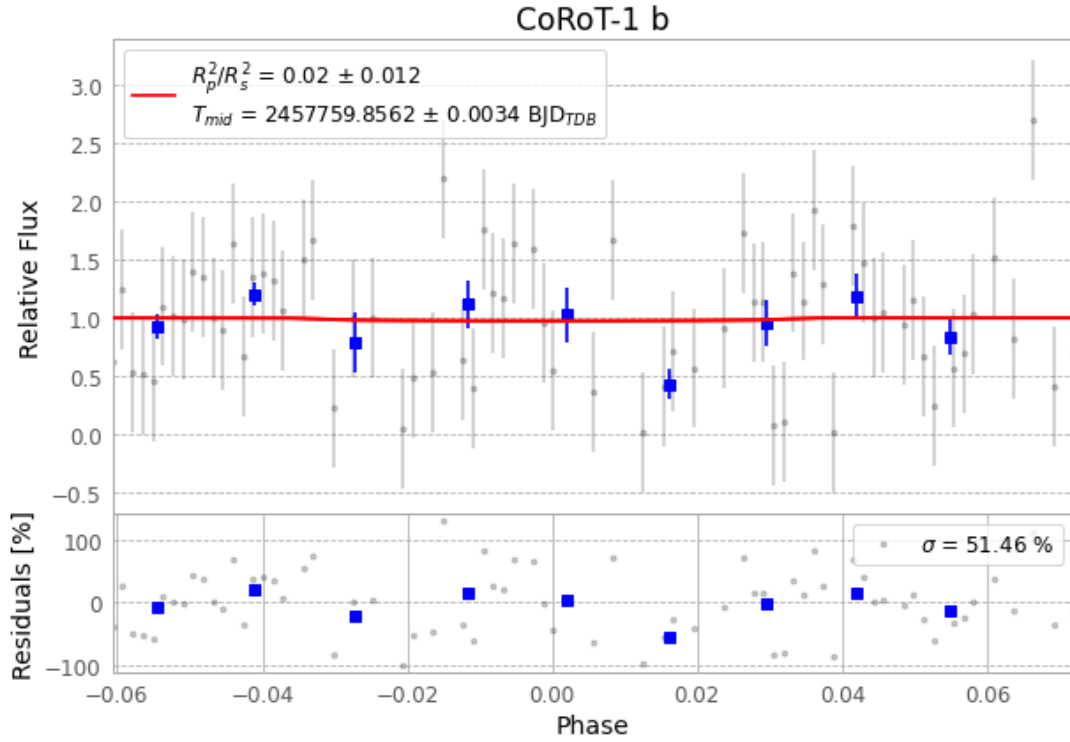


Figure 1 — FinalLightCurve_CoRoT-1 b_2017-01-05.png (SHA-256 8db5127231ea9190...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [61, 437], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 762b4bbc68ba8b0d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

97 FITS frames (64 MB), CoRoT-1170106061212.FITS → CoRoT-1170106110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-170106.log (04183e71ee93...), FinalLightCurve_CoRoT-1 b_2017-01-05.png (8db5127231ea...), FinalLightCurve_CoRoT-1 b_2017-01-05.csv (5bfbe14c8da2...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:55Z.